5 **(a)** Explain why the gravitational field strength is equal to the acceleration of free fall.

.....

.....

.....

..... [2]

(b) The mean radius of the Earth is 6370 km.

Show that the mass of the Earth is 5.97×10^{24} kg.

[1]

(c) Determine the gravitational potential at the surface of the Earth.

gravitational potential = ... J kg^{-1} [1]

(d) Explain why the gravitational potential is negative.

.....

.....

.....

..... [2]

[Total: 6]